

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Worked Solutions

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.**Solution**

1. Use $a^0 = 1$ and $a^{-n} = 1/a^n$; convert $0.2 = 1/5$.

Final answer

$$1, \frac{1}{8}, 25$$

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.**Solution**

1. Add exponents when multiplying, subtract when dividing, and multiply exponents for a power of a power.

Final answer

$$a^3, a^8, \frac{b^2}{a^2}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q003 **Written****Find:** (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.**Solution**1. Apply the zero and reciprocal-power laws; $0.25 = 1/4$.**Final answer** $1, \frac{1}{9}, 16$

Worked Solutions

Laws of Exponents and Roots

Q004 Written

Simplify the eight source expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Solution

1. Collect the exponent of each common base; distribute a negative outer power over every factor.

Final answer

$$a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q005 Written

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.**Solution**

1. Convert terminating decimals to fractions before applying the reciprocal rule.

Final answer $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Worked Solutions

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in source Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Solution

1. Use one exponent line for each common base and leave negative powers in reciprocal form when useful.

Final answer

$$a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$$

Worked Solutions

Laws of Exponents and Roots

Q007 Written

Simplify source Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3a^{-4} \div (2a)^{-2}$.

Solution

1. Distribute powers first, then combine exponents base by base; keep coefficients separate from variables.

Final answer

$$25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$$

Worked Solutions

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Solution

1. Rewrite the first factor as $(2a^2b^3)^{-2}$; distribute powers and combine all a- and b-exponents.

Final answer

$$\frac{1}{a^7b^6}$$

Q009 Written

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Solution

1. Recognise $64 = 4^3$ and $1/27 = (1/3)^3$.

Final answer

$$4, \frac{1}{3}$$

Worked Solutions

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[4]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Combine equal-index roots, factor perfect powers, then collect like cube-root terms.

Final answer

6, 5, 2, 3, $2\sqrt[3]{2}$

Q011 Written

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Solution

1. Use perfect fourth powers.

Final answer

3, $\frac{1}{2}$

Worked Solutions

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Solution

- Combine roots with the same index and extract perfect powers; leave the nested radical exact.

Final answer

4, 2, 10, $\sqrt[3]{4}$, $4\sqrt[3]{3}$

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Solution

- Prime-factorise each radicand and match the index to a perfect power.

Final answer

6, 5, 3, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{3}$, 1, 10, 0.1

Worked Solutions

Laws of Exponents and Roots

Q014 Written

Perform the fifteen source calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Solution

1. Convert every radicand to prime factors, combine equal-index roots, and collect only like radicals.

Final answer

$2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Worked Solutions

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Solution

1. Since $4096 = 2^{12}$, both numerator and denominator equal 2.

Final answer

1

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Solution

1. Use $\sqrt[n]{a^m} = a^{m/n}$ and move a radical denominator to a negative exponent.

Final answer

$5^{1/2}$, $2^{4/5}$, $3^{-3/5}$

Worked Solutions

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.**Solution**

1. Rewrite each base as a perfect power, multiply fractional exponents, and apply negative powers last.

Final answer

$$9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$$

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.**Solution**

1. Index over radicand exponent gives the rational power; reciprocals change the sign.

Final answer

$$a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(27/125)^{1/2}]^{-4/3}$.**Solution**

1. Prime-factorise the bases before multiplying the fractional exponents.

Final answer

$$27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$$

Q020 Written

Express the nine source expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.**Solution**

1. Translate each root and reciprocal directly into a rational exponent.

Final answer

$$2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve source expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/2)^{-1/3}]^{-2}$, $[(1/2)^{-1/3}]^{-2}$, $[(1/2)^{-1/3}]^{-2}$

Solution

1. Express every base as a power of a small integer, then multiply exponents exactly.

Final answer

$125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Worked Solutions

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Solution

1. Since $1000/27 = (10/3)^3$, the inner power is $(10/3)^{-2}$; multiply the remaining exponents.

Final answer

$$\frac{10}{3}$$

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[5]{32} \times \sqrt[3]{16}$.

Solution

1. Make the bases the same, then add/subtract rational exponents.

Final answer

1, 6, 6, 2

Worked Solutions

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Solution

1. Rewrite powers of 4 and 27 with bases 2 and 3, then combine exponents.

Final answer

5, 8, 7, 9

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Solution

1. Translate every radical to exponents of a and b, then collect like bases.

Final answer

$a^{1/3}$, $b\sqrt{a}$

Worked Solutions

Laws of Exponents and Roots

Q026 Written

Perform the twelve numerical calculations in source Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Solution

1. Make all bases equal before adding and subtracting rational exponents; extract perfect powers.

Final answer

$$1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$$

Q027 Written

For $a > 0, b > 0$, simplify the six source expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

Solution

1. Convert all radicals to rational powers and combine exponents separately for a and b.

Final answer

$$a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$$

Worked Solutions

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Solution

1. The printed decimal is $0.004 = 1/250$, so the exact result is not rational despite the prompt's wording; retain the exact radical form.

Final answer

$$\left(\frac{2025\sqrt{10}}{64} \right)^{3/2}$$

Q029 Written

Assess the statement $\sqrt{a^2} = a$ **for all real a, using positive and negative examples.**

Solution

1. For $a = 3$ the value is 3; for $a = -3$ the value is 3, not -3.

Final answer

Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Worked Solutions

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Solution**

1. Take the reciprocal of 2^3 .

Correct option: B**Final answer** $1/8$

Q031 MCQ

When multiplying equal bases, we:**Solution**

1. Use $a^m a^n = a^{m+n}$.

Correct option: B**Final answer**

add exponents

Worked Solutions

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:**Solution**1. Subtract: $5 - (-3) = 8$.

Correct option: B

Final answer a^8

Q033 MCQ

A negative exponent means:**Solution**1. Use $a^{-n} = 1/a^n$.

Correct option: B

Final answer

take the reciprocal power

Worked Solutions

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Solution**1. $0.04 = 1/25$, so the reciprocal square is 25^2 .

Correct option: C

Final answer

625

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Solution**1. Compute $-6 + 4 = -2$.

Correct option: A

Final answer

-2

Worked Solutions

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Solution**1. Combine the final powers as $a^{-7}b^{-6}$.

Correct option: B

Final answer

$$1/(a^7b^6)$$

Q037 MCQ

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:**Solution**1. The exponent is $5 - 6 + 1 = 0$.

Correct option: B

Final answer

$$1$$

Worked Solutions

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:**Solution**1. Since $4^3 = 64$.

Correct option: B

Final answer

4

Q039 MCQ

 $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Solution**1. Combine to $\sqrt[3]{216} = 6$.

Correct option: C

Final answer

6

Worked Solutions

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Solution**

1. $(1/2)^4 = 1/16$.

Correct option: A**Final answer**

$$1/2$$

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:**Solution**

1. Collect $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Correct option: B**Final answer**

$$2\sqrt[3]{2}$$

Worked Solutions

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Solution

1. $5^4 = 625$.

Correct option: B

Final answer

5

Q043 MCQ

When combining roots with the same index, multiply:

Solution

1. Use $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$.

Correct option: B

Final answer

the radicands

Worked Solutions

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Solution**

1. Both nested roots are 2.

Correct option: B**Final answer**

1

Q045 MCQ

 $\sqrt[5]{2^4}$ in rational-index form is:**Solution**

1. Index becomes the denominator.

Correct option: A**Final answer** $2^{4/5}$

Worked Solutions

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Solution**

1. $(3^3)^{2/3} = 3^2$.

Correct option: C**Final answer**

9

Q047 MCQ

A reciprocal square root is written as:**Solution**

1. A denominator changes the exponent sign.

Correct option: B**Final answer**

$$a^{-1/2}$$

Worked Solutions

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Solution**1. $81^{3/4} = 27$, then take the reciprocal.

Correct option: C

Final answer

1/27

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:**Solution**

1. Radical index is the denominator.

Correct option: A

Final answer

3/4

Worked Solutions

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:**Solution**

1. Use $64/27 = (4/3)^3$ and multiply exponents.

Correct option: C

Final answer

16/9

Q051 MCQ

The 6-3 challenge simplifies to:**Solution**

1. Collect the three rational exponents.

Correct option: B

Final answer

10/3

Worked Solutions

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3}2^{1/2} \div 2^{7/6}$ equals:**Solution**1. The exponent is $4/6 + 3/6 - 7/6 = 0$.

Correct option: B

Final answer

1

Q053 MCQ

 $4^{5/6}2^{4/3}$ equals:**Solution**1. Rewrite $4 = 2^2$, giving exponent 3.

Correct option: C

Final answer

8

Worked Solutions

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:**Solution**1. Compute $3/2 - 1/3 - 5/6 = 1/3$.

Correct option: A

Final answer

$$a^{1/3}$$

Q055 MCQ

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:**Solution**

1. The exponents add to 1.

Correct option: C

Final answer

$$6$$

Worked Solutions

Laws of Exponents and Roots

Q056 MCQ

The source expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Solution

1. Translate roots before combining.

Correct option: B

Final answer

combining rational exponents

Q057 MCQ

The printed decimal in the 6-4 challenge is:

Solution

1. Retain the ministry's printed value exactly.

Correct option: B

Final answer

0.004

Worked Solutions

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:

Solution

1. The principal square root is nonnegative.

Correct option: C

Final answer

$|a|$

Quiz MCQ 1 Concept Quiz · MCQ

Which law gives $a^m a^n = a^{m+n}$?

Solution

1. Add exponents for multiplication.

Correct option: A

Final answer

same-base multiplication

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Solution

1. Combine to $\sqrt[3]{125}$.

Correct option: B

Final answer

5

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The value of $64^{-3/2}$ is:

Solution

1. $64^{3/2} = 512$, then take the reciprocal.

Correct option: C

Final answer

1/512

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Solution

1. The principal root is $|a| = -a$ when a is negative.

Correct option: B

Final answer

$$-a$$

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $a^{-3}a^7 \div a^2$.

Solution

1. The exponent is $-3 + 7 - 2 = 2$.

Final answer

$$a^2$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Extract $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Final answer

$$2\sqrt[3]{2}$$

Quiz WRITTEN 7 Concept Quiz · Written

Evaluate $125^{-2/3}$.

Solution

1. $125^{2/3} = 25$, then take the reciprocal.

Final answer

$$1/25$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Solution

1. Use one positive and one negative counterexample.

Final answer

$\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}, \frac{10}{3}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 1, 9, 5, 9, $\sqrt{5}$, 4, 5, 3, 2, 1, 8, $\sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Worked Solutions

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Solution

1. Plot $x = -1, 0, 1$. A base > 1 gives growth; a base between 0 and 1 gives decay. The curve stays above the x-axis.

Final answer

Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.

Q002 Written

Draw $y = 2^x$ and $y = (1/2)^x$.

Solution

1. Use $(-1, 1/2), (0, 1), (1, 2)$ for 2^x and $(-1, 2), (0, 1), (1, 1/2)$ for its reciprocal-base graph.

Final answer

2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.

Worked Solutions

Exponential Functions and Their Applications

Q003 Written

Draw the six source graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Solution

1. For each graph mark $x = -1, 0, 1$, classify the base, and sketch a curve that never meets the x-axis.

Final answer

Bases 4, 10, 3/2: strictly increasing. Bases 2/3, 1/4, 1/10: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.

Q004 Written

Draw $y = \pi^x$ **for** $-1 \leq x \leq 2$, **marking the key points.**

Solution

1. Because $\pi > 1$, the curve increases. Restrict the drawing to the interval $[-1, 2]$ and keep $y > 0$.

Final answer

Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.

Worked Solutions

Exponential Functions and Their Applications

Q005 Written

Order the source sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Solution

1. For a base $0 < a < 1$, larger exponents give smaller values. For the second set rewrite with a common base 2 and compare exponents.

Final answer

$$0.7^2 < 0.7^{-1} < 0.7^{-3}; \text{ and } 5/64 < 5/8 < 6/16.$$

Q006 Written

Order the three displayed Try sets: 5^0 , 5^{-4} , $5^{3/2}$; $(1/3)^7$, $(1/3)^2$, $(1/3)^{-5}$; and 3^0 , 3^2 , 3^{-2} .

Solution

1. Make the base common. Keep exponent order for base > 1 and reverse it for $0 < a < 1$.

Final answer

$$5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2.$$

Worked Solutions**Exponential Functions and Their Applications****Q007** Written

Order the source sets with bases 3, 2, $\frac{3}{2}$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Solution

1. The nine source comparisons are solved by one rule: rewrite, state whether the base is above or below one, then write the ordered chain.

Final answer

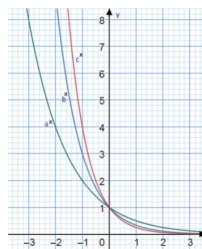
Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.

Worked Solutions

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a , b , c .



Solution

1. Convert radicals to rational powers before ordering; the opposite graph reverses the order when its base lies between 0 and 1.

Final answer

Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Worked Solutions

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Solution

1. Rewrite with a common base > 1 , compare exponents, and reverse an inequality only when the chosen base is between 0 and 1.

Final answer

$x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, **and the reciprocal-base inequalities.**

Solution

1. The Try set is source-faithfully solved by the same two-step method: common base, then exponent equation/inequality with sign control.

Final answer

Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .

Worked Solutions

Exponential Functions and Their Applications

Q011 Written

Solve all 24 source equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Solution

1. For each subpart: factor numerical constants into one base, write the exponent comparison, state $\text{base} > 1$ or $0 < \text{base} < 1$, and solve.

Final answer

Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Solution

1. Use base $4 > 1$, solve the quadratic inequality, then apply the physical time restriction $x \geq 0$.

Final answer

$x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.

Worked Solutions

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Solution

1. Set $t = a^x > 0$, solve the quadratic in t , discard non-positive roots, then convert back to x .

Final answer

First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Solution

1. The substitution changes the exponential problem to a quadratic; the positivity condition on t is mandatory before returning to x .

Final answer

Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.

Worked Solutions

Exponential Functions and Their Applications

Q015 Written

Solve the twelve source items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Solution

1. Write $t > 0$ beside the substitution, solve the quadratic, reject $t \leq 0$, and only then use monotonicity of the base to recover x .

Final answer

For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Solution

1. Factor the quadratic as $(t - 1)(t - 4) \leq 0$, convert the bounds with the decreasing base $1/2$, then solve the resulting quadratic inequalities.

Final answer

Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.

Worked Solutions

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Solution

1. Find the exact t -range first, compare the vertex with both endpoints, then convert t back to x .

Final answer

Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Solution

1. The quadratic has vertex $t = -1$, outside the positive t -range, so compare the two endpoints.

Final answer

Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.

Worked Solutions

Exponential Functions and Their Applications

Q019 Written

Solve the three source extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; **and the third displayed interval/model.**

Solution

1. A correct solution must include the t-range; an unconstrained quadratic vertex is not accepted unless it lies in that range.

Final answer

In each case set $t = a^x$, write the exact t-range, and compare the quadratic vertex and endpoints; report both extreme values and their x-locations.

Q020 Written

For $-1 \leq x \leq 2$, **optimise** $y = 2 \cdot 4^x - 2^{x+2} + 11$.

Solution

1. Use the interval $1/2 \leq t \leq 4$, evaluate the vertex $t = 1$, and compare both endpoints.

Final answer

Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.

Worked Solutions

Exponential Functions and Their Applications

Q021 **Written**

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Solution

1. AM-GM gives $a^x + a^{-x} \geq 2$, with equality at $a^x = 1$, i.e. $x = 0$; the two terms move in opposite directions away from zero.

Final answer

h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.

Q022 **MCQ**

Which base gives a strictly decreasing exponential graph?

Solution

1. A base between 0 and 1 gives decay.

Correct option: C

Final answer

$1/2$

Worked Solutions

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Solution**

1. Exponential values stay positive and approach zero.

Correct option: B**Final answer**

$$y = 0$$

Q024 MCQ

The range of $y = 5^x$ is:**Solution**

1. Every exponential output is positive.

Correct option: C**Final answer**

$$(0, \infty)$$

Worked Solutions

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?**Solution**1. At $x = 1$, $y = \pi$.

Correct option: B

Final answer $(1, \pi)$

Q026 MCQ

For $0 < a < 1$ and $r < s$, which is true?**Solution**

1. The exponent order reverses below base 1.

Correct option: B

Final answer $a^r > a^s$

Worked Solutions

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Solution**

1. Rewrite all terms using one base.

Correct option: B**Final answer**

make the bases equal

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Solution**

1. Base 3 is greater than 1.

Correct option: C**Final answer** 3^2

Worked Solutions

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Solution

1. $(2^{2/3})^3 = 2^2$.

Correct option: B

Final answer

$$2^2$$

Q030 MCQ

Solving $4^{x-1} = 8^{x+3}$ gives:

Solution

1. Use $2^{2x-2} = 2^{3x+9}$.

Correct option: A

Final answer

$$-11$$

Worked Solutions

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Solution**

1. Decreasing functions reverse inequalities.

Correct option: B**Final answer**

a reversed exponent order

Q032 MCQ

The solution of $3^x = 81$ is:**Solution**

1. $81 = 3^4$.

Correct option: C**Final answer**

4

Worked Solutions

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Solution

1. Solve $(x - 2)(x + 1) > 0$, then use $x \geq 0$.

Correct option: B

Final answer

$$x > 2$$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Solution

1. Every positive base power is positive.

Correct option: C

Final answer

$$t > 0$$

Worked Solutions

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Solution**1. Discard the negative root because $t > 0$.**Correct option:** B**Final answer**

$$(3 + \sqrt{73})/2$$

Q036 MCQ

The challenge quadratic in t is:**Solution**1. Set $t = (1/2)^{x^2 - 2x}$.**Correct option:** B**Final answer**

$$t^2 - 5t + 4$$

Worked Solutions

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Solution**

1. A power a^x cannot be zero or negative.

Correct option: C**Final answer**

$$t \leq 0$$

Q038 MCQ

For extrema, the first required range is the range of:**Solution**

1. The quadratic is constrained by the exact t-range.

Correct option: B**Final answer**

$$t = a^x$$

Worked Solutions

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Solution**1. Complete the square: $(t - 2)^2 - 4$.

Correct option: B

Final answer

-4

Q040 MCQ

The C06-K02 challenge has minimum value:**Solution**1. $y = 2(t - 1)^2 + 9$.

Correct option: A

Final answer $9atx = 0$

Worked Solutions

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Solution**1. AM–GM gives a minimum of 2 at $x = 0$.

Correct option: C

Final answer

2

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?**Solution**1. Equality in AM–GM occurs when $a^x = 1$.

Correct option: B

Final answer $x = 0$

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Solution

1. The domain is all real numbers and the range is positive.

Correct option: A

Final answer

range $(0, \infty)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For base 0.5, increasing the exponent makes the value:

Solution

1. The function is decreasing.

Correct option: B

Final answer

smaller

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The positive-substitution condition is:

Solution

1. An exponential power is always positive.

Correct option: C

Final answer

$$t > 0$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $y = 2(t - 1)^2 + 9$, the minimum is:

Solution

1. The square is zero at $t = 1$.

Correct option: C

Final answer

$$9$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 5 Concept Quiz · Written

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Solution

1. Use the three points and the base $0 < 1/4 < 1$.

Final answer

Decreasing; $D = \mathbb{R}, R = (0, \infty), y = 0$

Quiz WRITTEN 6 Concept Quiz · Written

Order $2^{-3}, 2^0, 2^{5/2}$.

Solution

1. Base 2 is greater than 1, so preserve exponent order.

Final answer

$$2^{-3} < 2^0 < 2^{5/2}$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $4^x - 5 \cdot 2^x + 4 = 0$.**Solution**1. Let $t = 2^x > 0$: $t^2 - 5t + 4 = (t - 1)(t - 4) = 0$.**Final answer**

$$x = 0 \text{ or } x = 2$$

Quiz WRITTEN 8 Concept Quiz · Written**Find the minimum of** $a^x + a^{-x}$ **for** $a > 1$.**Solution**1. By AM-GM, $a^x + a^{-x} \geq 2$, with equality when $a^x = 1$.**Final answer**

$$2 \text{ at } x = 0$$

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q001 **Written****Convert the source forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.****Solution**

1. Use $a^p = M \iff \log_a M = p$; swap the last two numbers.

Final answer

$$\log_2 32 = 5; 3^2 = 9$$

Q002 **Written****Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.****Solution**

1. Identify base, exponent, and argument; then interchange the exponent and argument positions.

Final answer

$$\log_2 64 = 6; \log_4(1/16) = -2, \text{ with the same swap rule for the remaining items.}$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q003 Written

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Solution

1. Check $a > 0$, $a \neq 1$, $M > 0$, then write the equivalent form without changing the values.

Final answer

Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 Written

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

Solution

1. First find x from $5^x = 25$, then recognise $2 = (1/2)^{-1}$.

Final answer

$x = 2$, so $\log_{1/2} 2 = -1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1, \log_5 5, \log_2 8, \log_2 16, \log_5(1/5), \log_7(49^{3/2})$.

Solution

1. Rewrite each argument as a power of its base; the exponent is the logarithm.

Final answer

Values: 0, 1, 3, 4, -1 , $3/2$.

Q006 Written

Find the six Try values $\log_5 1, \log_3 3, \log_4 4, \log_3 27, \log_2(1/4), \log_5(1/125)$.

Solution

1. Use $\log_a 1 = 0$, $\log_a a = 1$, and the power definition.

Final answer

0, 1, 1, 3, -2 , -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q007 Written

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

Solution

1. Do not approximate. For reciprocals, use a negative exponent; for roots, use a fractional exponent.

Final answer

Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

Q008 Written

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

Solution

1. Since $a^2 = (a^3)^{2/3}$, the logarithm is $2/3$.

Final answer

$2/3$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.**Solution**

1. Combine products for addition and quotients for subtraction, then express the result as a base power.

Final answer

1 and 1.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.**Solution**

1. Move coefficients into exponents first, combine logs, simplify the argument, and read the final exponent.

Final answerEach expression reduces to one logarithm and then an integer; the displayed worked example gives -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q011 Written

Perform all 18 source expressions (a)+(r), combining same-base logarithms exactly.**Solution**

1. Apply the power law before product/quotient combination; simplify the final argument as a power of the base.

Final answer

Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.**Solution**

1. Move coefficients inside, combine the argument, and reduce it to a power of 5.

Final answer

1.

Worked Solutions

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ **and** $\log_{10} 3 = b$, **express** $\log_{10} 24, \log_{10}(27/4), \log_{10} 45, \log_{10}(24/75)$ **in terms of** a, b .

Solution

1. Prime-factorise the argument, use $5 = 10/2$ when required, then apply product, quotient, and power laws.

Final answer

$3a + b, 2 - 2a + 3b, 2a + 1 - a, b + 2 - 2a.$

Q014 Written

Given $\log_{10} 2 = a, \log_{10} 3 = b$, **express the four Try arguments** $72, 3^8/10, 15, 12$ **in terms of** a, b .

Solution

1. Write every integer as $2^m 3^n 5^p$, replace $\log_{10} 2, \log_{10} 3$, and convert the 5-factor using $1 - a$.

Final answer

Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

Worked Solutions

Introduction to Logarithms and Their Properties

Q015 Written

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Solution

1. Keep numerator and denominator factors separate until the final subtraction.

Final answer

Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 Written

If $\log_c 3 = x, \log_c 2 = y$, write $\log_c 12, \log_c(3/8), \log_c \sqrt{6}, \log_2 3, \log_{3b}(2^b c)$ in terms of x, y .

Solution

1. Factorise arguments, use the logarithm laws, and use change of base for the last two forms.

Final answer

$x + 2y, x - 3y, (x + y)/2, x/y, (1 + by)/(bx).$

Worked Solutions

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Solution

1. Use $\log_a b = \log_c b / \log_c a$, choosing a base that turns both numbers into powers.

Final answer

$3/2$ and -3 .

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Solution

1. Choose the smallest convenient base; do not use decimal approximations when exact powers are available.

Final answer

Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

Worked Solutions

Introduction to Logarithms and Their Properties

Q019 Written

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.**Solution**

1. Convert numerator and denominator to the same base, then cancel common factors.

Final answer

All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 Written

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.**Solution**

1. Apply change of base to 2, then express the radical denominator as a power.

Final answer

$$\frac{\log_2 3 - 5/2}{2 \log_2 3}.$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q021 **Written****Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.****Solution**

1. Make all logarithms use base 2 (or 3), then cancel adjacent change-of-base factors.

Final answer

Use reciprocal and power identities; the worked values are exact constants.

Q022 **Written****Solve the four Try products of logarithms after making the bases the same.****Solution**

1. Treat each logarithm as a change-of-base fraction and cancel before multiplying.

Final answer

Each product collapses to a rational number after change of base and cancellation.

Worked Solutions

Introduction to Logarithms and Their Properties

Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.**Solution**

1. Make the base 2, 3, or 5 consistently, simplify inside parentheses, then multiply.

Final answer

Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 Written

Simplify $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ for $x > 1$.**Solution**

1. Convert each factor to a common natural-log ratio; exponents contribute $1/2, 3/2, 8$, whose product is 12.

Final answer

12.

Worked Solutions

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.**Solution**

1. Use $\log_4(x^2) = 2\log_4 |x|$; negative x belongs to f 's domain but not g 's.

Final answer

They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.**Solution**

1. Plot $y = -1, 0, 1$ in the exponential counterpart, then reflect in $y = x$.

Final answer

Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

Worked Solutions

Introduction to Logarithms and Their Properties

Q027 **Written****State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.****Solution**

1. Use the domain $x > 0$, vertical asymptote $x = 0$, and inverse reflection.

Final answer

Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 **Written****Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.****Solution**

1. Every graph passes through $(1, 0)$; do not draw it across the y-axis.

Final answer

Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

Worked Solutions

Introduction to Logarithms and Their Properties

Q029 Written

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Solution

1. Translate the standard inverse pair by the same horizontal/vertical shifts and identify the resulting symmetry line.

Final answer

They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 Written

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2 \log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Solution

1. Compare arguments only after the bases are the same and the base condition is stated.

Final answer

Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q031 **Written****Order the four Try sets of logarithms using monotonicity and change of base.****Solution**

1. Rewrite $k \log_a M$ as $\log_a (M^k)$, then compare the arguments.

Final answer

Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 **Written****Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.****Solution**

1. Move coefficients inside first, then compare arguments with the correct direction.

Final answer

Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

Worked Solutions

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Solution

1. Express both logs with a common base and state the monotonicity case before writing the inequality.

Final answer

Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 MCQ

$\log_2 32$ equals:

Solution

1. Since $2^5 = 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:**Solution**

1. $(1/2)^{-1} = 2.$

Correct option: B

Final answer

-1

Q036 MCQ

 $\log_5(1/125)$ equals:**Solution**

1. $1/125 = 5^{-3}.$

Correct option: A

Final answer

-3

Worked Solutions

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:**Solution**1. Write $49^{3/2} = 7^3$.

Correct option: C

Final answer $3/2$

Q038 MCQ

 $\log_a M + \log_a N$ equals:**Solution**

1. Use the product law.

Correct option: B

Final answer $\log_a(MN)$

Worked Solutions

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Solution**1. Use $k \log_a M = \log_a (M^k)$.

Correct option: B

Final answer

$$\log_5 2^3$$

Q040 MCQ

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:**Solution**1. $24 = 2^3 \cdot 3$.

Correct option: C

Final answer

$$3a + b$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Solution

1. Use $\sqrt{6} = 6^{1/2}$.

Correct option: B

Final answer

$$(x + y)/2$$

Q042 MCQ

$\log_{16} 64$ equals:

Solution

1. $16 = 2^4, 64 = 2^6$.

Correct option: B

Final answer

$$3/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Solution**

1. Use the quotient of common-base logs.

Correct option: B**Final answer**

$$\log_c b / \log_c a$$

Q044 MCQ

 $\log_4 16 \cdot \log_9 3$ equals:**Solution**1. Values are $2 \cdot 1/2$.**Correct option:** A**Final answer**

$$1/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Solution**

1. Multiply exponent factors after change of base.

Correct option: C**Final answer**

12

Q046 MCQ

The domain of $y = \log_a x$ is:**Solution**

1. The argument must be positive.

Correct option: C**Final answer** $x > 0$

Worked Solutions

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Solution

1. Inverse graphs reflect in $y = x$.

Correct option: C

Final answer

$$y = x$$

Q048 MCQ

For $a > 1$ and $M < N$, which is true?

Solution

1. The logarithm is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Solution**

1. The logarithm is decreasing.

Correct option: A**Final answer**

$$\log_a M > \log_a N$$

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Solution**1. Logarithmic graphs require $x > 0$.**Correct option:** A**Final answer**

$$x = 0$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Solution

1. Move the coefficient into the exponent.

Correct option: A

Final answer

$$\log_a M^{-2}$$

Q052 MCQ

The base of $\log_a M$ is:

Solution

1. The subscript is the base.

Correct option: B

Final answer

$$a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Solution**

1. The number inside the logarithm is the argument.

Correct option: B**Final answer** M

Q054 MCQ

 $\log_2(1/8)$ equals:**Solution**

1. $1/8 = 2^{-3}$.

Correct option: A**Final answer** -3

Worked Solutions

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:**Solution**

1. The argument equals the base.

Correct option: B

Final answer

1

Q056 MCQ

 $\log_3 1$ equals:**Solution**

1. Every allowed base to power zero is 1.

Correct option: B

Final answer

0

Worked Solutions

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:**Solution**

1. Use the quotient law.

Correct option: B

Final answer

$$\log_a(M/N)$$

Q058 MCQ

 $2\log_3 5$ equals:**Solution**

1. Move 2 into the argument as an exponent.

Correct option: B

Final answer

$$\log_3 25$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Solution

1. $12 = 2^2 \cdot 3$.

Correct option: B

Final answer

$$2a + b$$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Solution

1. $5 = 10/2$.

Correct option: B

Final answer

$$1 - a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:**Solution**1. $8 = 2^3, 32 = 2^5$.

Correct option: B

Final answer $5/3$

Q062 MCQ

 $\log_{25} 125$ equals:**Solution**1. $25 = 5^2, 125 = 5^3$.

Correct option: B

Final answer $3/2$

Worked Solutions

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:**Solution**

1. Change of base makes reciprocal factors.

Correct option: B

Final answer

1

Q064 MCQ

 $\log_2 8 \cdot \log_8 2$ equals:**Solution**

1. The factors are reciprocals.

Correct option: B

Final answer

1

Worked Solutions

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Solution**

1. Both numbers are powers of 2.

Correct option: A**Final answer**

2

Q066 MCQ

The range of $y = \log_a x$ is:**Solution**

1. Every real exponent is attained.

Correct option: B**Final answer** $\mathbb{R}$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Solution

1. Swap the exponent and argument.

Correct option: A

Final answer

$$\log_a M = p$$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\log_2 8 + \log_2 4$ equals:

Solution

1. Combine to $\log_2 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 3 [Concept Quiz · MCQ](#)

$\log_{16} 64$ equals:

Solution

1. Use base 2 exponents 6/4.

Correct option: B

Final answer

$$3/2$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $a > 1$, $M < N$ implies:

Solution

1. The logarithmic function is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written**Find** $\log_3(1/27)$.**Solution**

1. $1/27 = 3^{-3}$.

Final answer -3 **Quiz WRITTEN 6** Concept Quiz · Written**Simplify** $\log_5 25 + \log_5 5$.**Solution**

1. Combine to $\log_5 125 = 3$.

Final answer 3

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 Concept Quiz · Written**Find $\log_{16} 64$ using change of base.****Solution**

1. $\log_2 64 / \log_2 16 = 6/4 = 3/2.$

Final answer

$$3/2$$

Quiz WRITTEN 8 Concept Quiz · Written**State the domain and range of $y = \log_a x$.****Solution**

1. The argument is positive and every real exponent is attained.

Final answer

Domain $x > 0$; range all real numbers

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4 (1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q034** C
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** B
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1** A**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** B**Quiz WRITTEN 5** -3 **Quiz WRITTEN 6** 3**Quiz WRITTEN 7** $\frac{3}{2}$ **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Worked Solutions

Applications of Logarithms

Q001 Written

Solve $\log_2 x = 4$.**Solution**

1. Set the argument equal to 2^4 ; hence $x = 16$.

Final answer

16

Q002 Written

Solve $\log_2(2x - 1) = \log_2 3$.**Solution**

1. Domain: $2x - 1 > 0$. Equal arguments give $2x - 1 = 3$, so $x = 2$.

Final answer $x = 2$

Worked Solutions

Applications of Logarithms

Q003 Written

Solve all 12 source logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Solution

1. Write every argument condition, combine logs, solve the resulting algebraic equation, and verify each candidate.

Final answer

Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.

Q004 Written

Solve $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ completely.

Solution

1. Domain gives $x > 2$. Convert $\log_9$ to base 3, solve $(x + 2)^2 / (x - 2)^2 = 9$, and retain only $x = 4$.

Final answer

$x = 4$

Worked Solutions

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.**Solution**

1. State the domain, preserve the sign for base 3, reverse it for base $1/2$, then intersect.

Final answer

$$0 < x < 81; \quad 5 < x < 6$$

Q006 Written

Solve the six source logarithmic inequalities using base monotonicity and argument conditions.**Solution**

1. For $a > 1$ keep the inequality direction; for $0 < a < 1$ reverse it before solving.

Final answer

Each solution is the intersection of the argument domain with the monotonicity interval.

Worked Solutions

Applications of Logarithms

Q007 Written

Solve all 24 source logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Solution

1. Never omit the positive-argument condition; it can remove one branch of an algebraic solution.

Final answer

Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.

Q008 Written

Solve the source challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Solution

1. Combine the logarithms, solve the rational inequality, and retain only the common domain.

Final answer

Reverse the comparison after converting to a base below one, then intersect with both argument conditions.

Worked Solutions

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0$.**Solution**

1. Let $t = \log_2 x$: $(5t - 1)(t - 3) < 0$, so $1/5 < t < 3$, then back-substitute.

Final answer

$$2^{1/5} < x < 8$$

Q010 Written

Solve the three source substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.**Solution**

1. The substitution turns the problem into a quadratic; only then return to the original logarithm.

Final answerSubstitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.

Worked Solutions

Applications of Logarithms

Q011 Written

Solve the source groups: three equations, three inequalities, and three exact logarithmic values.

Solution

1. Record the domain before setting $t = \log_a x$, and verify every returned value.

Final answer

All groups follow the same substitution, domain, solve, and back-substitution sequence.

Q012 Written

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x .

Solution

1. Let $t = \log_3 x$: $(t - 3)(t + 2) < 0$, so $-2 < t < 3$, hence $3^{-2} < x < 3^3$.

Final answer

$1/9 < x < 27$

Worked Solutions

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Solution

1. Let $t = \log_3 x$, so $0 \leq t \leq 3$; complete the square $y = 7 - (t - 2)^2$.

Final answer

Maximum 7 at $x = 9$; minimum 3 at $x = 1$.

Q014 Written

Find the maximum and minimum of the source logarithmic quadratic on its stated interval, with the corresponding values of x .

Solution

1. Convert the x -interval to a t -interval before reading the quadratic range.

Final answer

Use the vertex in $t = \log_a x$ and compare it with both endpoints.

Worked Solutions

Applications of Logarithms

Q015 Written

Solve the three source extrema problems, each with a different base and interval.

Solution

1. The maximum/minimum in t must be translated back to the exact $x = a^t$.

Final answer

For every item, map the domain to t , locate the vertex, and check endpoints.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Solution

1. Here $0 \leq t \leq 3$ and $y = 5 - (t - 1)^2$; compare $t = 0, 1, 3$.

Final answer

Maximum 5 at $x = 2$; minimum 4 at $x = 1$.

Worked Solutions

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Solution

1. Take common logs: $30(\log 2 + \log 3) = 23.343$ and $50(\log 2 - \log 3) = -8.805$.

Final answer

24 digits; 9th decimal place.

Q018 Written

Complete the source digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Solution

1. Use the larger absolute exponent for a number between two negative powers of ten.

Final answer

Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.

Worked Solutions

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Solution

1. Compute $50 \log_{10} 12 = 50(2 \log 2 + \log 3) = 53.965$; then use the characteristic and mantissa.

Final answer

55 digits; leading digit 1.

Q020 Written

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x .

Solution

1. Domain: $0 < x < 4, x \neq 1$. For $0 < x < 1$ reverse the sign and for $1 < x < 4$ keep it; both cases are impossible.

Final answer

No real solution.

Worked Solutions

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Solution**

1. A logarithm argument must be positive.

Correct option: C**Final answer**

$$x > 5$$

Q022 MCQ

If $\log_2(2x - 1) = \log_2 3$, then $x =$:**Solution**

1. Equate same-base arguments.

Correct option: B**Final answer**

$$2$$

Worked Solutions

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Solution**

1. The domain $x > 2$ rejects the other algebraic root.

Correct option: C**Final answer**

4

Q024 MCQ

For same-base logarithms, equality of logs gives:**Solution**

1. The logarithm is one-to-one on its domain.

Correct option: A**Final answer**

arguments equal

Worked Solutions

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:**Solution**

1. The logarithm is increasing.

Correct option: B**Final answer**

$$M < N$$

Q026 MCQ

For base $0 < a < 1$, the inequality direction is:**Solution**

1. A reciprocal-base logarithm is decreasing.

Correct option: B**Final answer**

reversed

Worked Solutions

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Solution**

1. The argument must be positive.

Correct option: C

Final answer

$$x > 5$$

Q028 MCQ

The warm-up intersection is:**Solution**1. Intersect the algebraic interval with $x > 5$.

Correct option: B

Final answer

$$5 < x < 6$$

Worked Solutions

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Solution**

1. It turns the logarithm into a single variable.

Correct option: B**Final answer**

$$t = \log_a x$$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:**Solution**

1. Factor and use the sign chart.

Correct option: B**Final answer**

$$1/5 < t < 3$$

Worked Solutions

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Solution**

1. A logarithm value can be any real number.

Correct option: B**Final answer**

$$t \in \mathbb{R}$$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:**Solution**

1. Convert $-2 < t < 3$ back with base 3.

Correct option: B**Final answer**

$$1/9 < x < 27$$

Worked Solutions

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Solution

1. The vertex gives the maximum.

Correct option: C

Final answer

$$t = 2$$

Q034 MCQ

The correct first step for extrema is:

Solution

1. Then map the domain to a t-range.

Correct option: B

Final answer

$$\text{set } t = \log_a x$$

Worked Solutions

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Solution**1. Apply $\log_3$ to the endpoints.

Correct option: B

Final answer $[0, 3]$

Q036 MCQ

The water-tank challenge maximum health rating is:**Solution**1. The vertex is at $t = 1$, so $x = 2$.

Correct option: C

Final answer

5

Worked Solutions

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Solution**

1. Use the characteristic of the common logarithm.

Correct option: B**Final answer**

$$[\log_{10} N] + 1$$

Q038 MCQ

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:**Solution**

1. Use the larger absolute negative exponent.

Correct option: B**Final answer** 9^{th}

Worked Solutions

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Solution**

1. A common logarithm has base 10.

Correct option: C

Final answer

10

Q040 MCQ

The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:**Solution**

1. The increasing and decreasing cases both contradict their domains.

Correct option: D

Final answer

empty

Worked Solutions

Applications of Logarithms

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Solve $\log_5(x - 1) = 2$.

Solution

1. $x - 1 = 25$.

Correct option: A

Final answer

$$x = 26$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $0 < a < 1$ **and** $M < N$, **compare** $\log_a M$ **and** $\log_a N$.

Solution

1. The function is decreasing.

Correct option: B

Final answer

$$\log_a M > \log_a N$$

Worked Solutions

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.

Solution

1. Let $t = \log_2 x$: $(4t - 1)(t - 1) = 0$.

Correct option: C

Final answer

$$x = 2, \sqrt{2}$$

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ **on** $1 \leq x \leq 27$.

Solution

1. Let $t = \log_3 x \in [0, 3]$; vertex at $t = 1$.

Correct option: D

Final answer

$$5 \text{ at } x = 3$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.**Solution**

1. $x + 1 = 9$.

Final answer

$$x = 8$$

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.**Solution**

1. Base 2 is increasing.

Final answer

$$x \geq 8$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $(\log_5 x)^2 - 4 < 0$.**Solution**

1. $-2 < \log_5 x < 2$.

Final answer

$$1/25 < x < 25$$

Quiz WRITTEN 8 Concept Quiz · Written**How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?****Solution**

1. $20(0.3010 + 0.4771) = 15.562$.

Final answer

16 digits

Answer key

Use the question number shown on each page.

- Q001** 16
Q002 $x = 2$
Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
Q004 $x = 4$
Q005 $0 < x < 81$; $5 < x < 6$
Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
Q009 $2^{1/5} < x < 8$
- Q010** Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
Q012 $1/9 < x < 27$
Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
Q017 24 digits; 9th decimal place.
Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019** 55 digits; leading digit 1.
Q020 No real solution.
Q021 C
Q022 B
Q023 C
Q024 A
Q025 B
Q026 B
Q027 C
Q028 B
Q029 B
Q030 B
Q031 B
Q032 B
Q033 C
Q034 B
- Q035** B
Q036 C
Q037 B
Q038 B
Q039 C
Q040 D
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 D
Quiz WRITTEN 5 $x = 8$
Quiz WRITTEN 6 $x \geq 8$
Quiz WRITTEN 7 $1/25 < x < 25$
Quiz WRITTEN 8 16 digits